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Optimizing sound diffusion in a concert hall using scale-model measurements and simulations

Hyun In Jo, Jin Yong Jeon *

Department of Architectural Engineering, Hanyang University, Seoul, 04763, South Korea

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ABSTRACT

To propose optimal guidelines for the diffusion design of concert halls, sound field variations in the presence or absence of reflectors (canopies and clouds) and diffusers were comprehensively examined in terms of stage and auditorium acoustics using 1:25 scale-model measurements and acoustic simulations. The sound field variations were investigated based on the just-noticeable differences. For stage acoustics, the stage support parameters ST_{Early} and ST_{Late} were analyzed; for auditorium acoustics, the reverberation time (RT), early decay time (EDT), clarity (C_{80}), and center time (T_S) were analyzed. The canopies reduced the reverberations but increased ST_{Early} and ST_{Late}. The clouds increased ST_{Early} and reduced the deviation of the auditorium EDT and T_S whilst increasing the initial EDT that had been reduced by the canopies. The diffusers decreased ST_{Early}, ST_{Late}, RT, EDT, and T_S regardless of the attachment position, and they increased the C_{80} of some auditoriums. However, diffuser attachments reduced the seat-to-seat deviations of acoustic parameters. Furthermore, attaching diffusers near to (as opposed to far from) the sound source represents the optimal design strategy for increasing the density of early reflections. This study is significant as an initial attempt to examine the effects of concert hall diffusion design in terms of both auditorium and stage acoustics, and it is significant in that it presents an appropriate method for evaluating the effects of a concert hall diffusion design that can be employed by architects and sound consultants. The findings of this study are expected to be suitable as basic data for the diffusion design of rectangular concert halls.

1. Introduction

Sound diffusion is an important factor in the sound environment design of various spaces within buildings. It is considered particularly important in music venues such as concert halls and opera houses. In general, diffusion design is used to reduce acoustic defects such as acoustic glare, echo problems, and tone coloration [1,2], and it has positive effects on spaciousness, acoustic quality, and the uniformity of acoustic parameter distributions [3,4]. However, several researchers claim that diffusion designs disperse the direct sound generated by performers on the stage, thereby reducing the sound presence and proximity and potentially affecting the desirable characteristics of music venues [5,6]. Such diffusion designs can be found in interior finishes with irregular surfaces and statues in traditional 19th century rectangular concert halls [7]. In modern halls (from the 20th century onwards), repeated periodic patterns of surface shapes are utilized [8]. Various other factors can also produce diffusion in modern halls, including ceiling shapes, balcony fronts, and reflectors. Therefore, to maximize the aforementioned positive effects of sound diffusion, it is necessary to propose an appropriate diffusion design from the early stages of music venue planning.

* Corresponding author. Dept. of Architectural Eng., Hanyang University , Seoul 04763, South Korea.
E-mail address: jyjeon@hanyang.ac.kr (J.Y. Jeon).

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In general, music venue acoustics can be classified as auditorium acoustics and stage acoustics. First, with regard to diffusion research for auditorium acoustics, studies into the quantification, design, and evaluation of diffusion performance have been conducted from various perspectives [9–11]. Representative quantitative evaluation indices for diffusion performance include the scattering coefficient (Sc) and diffusion coefficient (Dc), which were established by ISO 17497–1 [12] and ISO 17947–2 [13], respectively. In addition, various indices have been proposed to quantify diffusion performance. The standard deviation of the reverberation time (RT) in an auditorium has been proposed as an evaluation index [14]. Using the interaural cross-correlation (which varies significantly following the installation of a cylindrical diffuser inside the music venue) as a diffusion index has also been suggested [15]. The scatter-to-absorption ratio, mean scatter time, and diffusion time have also been proposed as diffusion performance evaluation indices for spaces [16]. Focusing on the fact that diffusion influences the acoustic parameter distributions, the relative standard deviation (RSD) of each acoustic parameter has been proposed as an evaluation index to quantify seat-to-seat deviation [17]. The number of peaks (Np), which quantifies the number of reflection peaks generated by diffusion, has also been suggested [18].

Using these indices, acoustic simulations and scale models have been used to predict the diffusion performance of music venues. In particular, because acoustic simulations have been actively used to predict acoustic performance in acoustic materials and structural analysis fields (e.g., stud, metamaterial, metal-framed walls, etc.) besides that of design, its applicability in the design of concert halls is very wide [19–21]. Geometric acoustic (GA) simulation software has been used to show that the early decay time (EDT) and clarity (C_{80}) of an auditorium increase and decrease, respectively, when the Sc values of its interior walls and ceiling increase from 0.1 to 0.9 [22]. Furthermore, the case of a diffuser surface shape that was modeled in such detail as to be highly similar to the real surface was compared with that of applying only the scattering coefficient value numerically to a flat surface. The former case produced results more similar to the auditory characteristics of actual diffusion [23]. It has also been found through acoustic simulation that acoustic parameters are influenced by the source-to-receiver distance [24]. However, Jeon et al. [11] increased the Dc for the wall and ceiling of an auditorium (excluding the stage and balcony) from 0.05 to 0.7 under identical sound absorption conditions in music venues of 12 shapes (e.g., fan shape and shoebox); they found that the results for most acoustic parameters were below the just-noticeable difference (JND), and that no clear change patterns could be generalized. This confirmed the limitation of GA simulation software. To address the limitations of GA simulation software, scale models have been considered in the design stage, and models with scales ranging from 1:8 to 1:50 have been found appropriate [25]. When 3D diffusers were attached to the walls around the stage and balcony fronts in a 1:10 scale model of a music venue, the RT and EDT of the auditorium and C_{80} of the area around the stage increased, whereas C_{80} of the back rows decreased [26]. In a follow-up study [11], 2D diffusers were attached to the same positions inside a 1:10 scale model of a concert hall; it was found that the RT always decreased in the auditorium; however, close to the sound source, the EDT increased and C_{80} decreased, whereas far from the sound source, the EDT decreased and C_{80} increased. In a 1:25 scale model, the RT and EDT decreased when the number of diffusers on the sidewall was increased; however, C_{80} did not show any distinct trend [27]. Other studies reported that the RT decreased but the EDT increased under the application of diffusers in 1:25 scale models [28]. Thus, most studies into the diffusion design of auditoriums have only examined changes in acoustic parameters therein according to the shapes or attachment positions of diffusers. However, because diffusers inside music venues can markedly influence the on-stage performance as well as the audio perceptions of the audience, their complex mutual effects must be examined.

The stage is critical in a music venue because it serves as a performance space and is the source of the sound that the audience hears. Most studies on stage acoustics have sought to strike a balance between the ensemble and perceived reverberance at which the performers can listen to the sounds of their fellow performers whilst also listening to their own sound [29–31]. Therefore, when designing a stage, various factors such as the shape, size, and height of the stage enclosure [including the walls around the stage and the reflector (canopy)] should be considered [25]. Representative quantitative evaluation indices for stage acoustics include the stage support parameter (ST), which is calculated as the ratio between the energies of the direct and reflected sound. The ST was established as a standard in ISO 3382–1 [32] because of its verifiable utility as an evaluation index in actual halls. To evaluate the stage acoustics based on this index, GA simulations and scale models are often used. First, from the perspective of the performers, it was found in a 1:25 scale model and GA simulation that when the stage ceiling height decreased, the ST increased; furthermore, the ST on the stage increased in certain cases in which diffusers were attached to the side and rear walls of the stage [33]. A study in which a 1:10 scale model and GA simulation were evaluated showed that installing detached reflectors whilst decreasing the stage volume, splay of the stage envelope, and stage width improved the initial energy on the stage [34]. However, when the initial energy was extremely large, the loudness decreased in the auditorium because performers tended to play at low volumes [35]. Therefore, it is recommended to install a canopy of 7–13 m to ensure −12 to −15 dB, which is a suitable range for the ST [2]. However, reflectors should be configured into sub-reflectors rather than on a single plane, owing to the ensemble of performers on the stage [36]. One study demonstrated that a narrow, high stage enclosure is more desirable for an orchestral concert, by measuring an actual music venue on a 1:1 scale [37]. Furthermore, a method of tilting, diffusive, and absorbent processing of the auditorium rear wall was also proposed to resolve the long-path echo generated during performances [38]. From the audience perspective, it was shown that splaying of the stage sidewalls and ceiling toward the auditorium can reinforce the late energy of the auditorium around the stage. It is also understood that a maximum stage width of 16 m and maximum stage depth of 11 m are appropriate when the auditorium acoustics are considered [3, 39]. Furthermore, when the music venue is designed to reduce the stage volume ratio, the RT and EDT may decrease slightly; however, the bass ratio of the auditorium is reinforced [40]. In addition, in terms of diffusion design, installing a pipe organ with Sc = 0.5 on the sidewall of the auditorium can increase the energy ratio of the sidewall reflections thereto [41].

As described above, stage acoustics studies have mostly focused on the effects of design factors on the performers on stage, whilst the effects of stage design factors on the auditorium remain under-researched. Moreover, even though many stage design factors can produce diffusion effects, few studies have examined them.

Studies into the diffusion design of music venues can be classified into those focusing on stage acoustics or auditorium acoustics;

however, few studies have examined their complex mutual effects in sufficient detail. Therefore, in this study, the diffusion effects of each design factor upon the stage and auditorium acoustics were analyzed using a GA simulation and scale model, with a focus on reflectors (which can produce a diffusion effect among the stage design factors) and diffusers (which are the most frequently used auditorium design factors). Based on the analysis results, guidelines for the optimal diffusion design are presented. This study is significant as the first research attempt to consider sound field variations in a concert hall, by applying diffusion design from the perspectives of both auditorium and stage acoustics. This is because few studies have been conducted on the influence of stage design factors (including ceiling reflectors) on the auditorium audience from a diffusion perspective, and studies that examine the effects of auditorium diffusion design factors (including walls) on stage performers are also insufficient. This study will play a potential role in allowing the architects, sound consultants, and conductors involved in concert hall design to suggest design directions to be pursued (from both audience and performer perspectives) for diffusion design in the initial concert-hall design stages. In addition, two representative methods (simulation and scale model) for evaluating concert hall acoustics during the concert hall design stage were compared, and the limitations and applications of each method were presented so that they can be used as basic data for concert hall design projects in the future.

The research questions addressed in this study are as follows:

- How do the sound fields on the stage and in the auditorium vary according to the reflector type (canopy and/or cloud)? Are the diffusion effects significant?
- How do the sound fields on the stage and in the auditorium vary according to the diffuser attachment position? Are the diffusion effects significant?
- How can reflectors and diffusers be utilized for the optimal diffusion design of concert halls?
- How does a scale model differ from a GA simulation in concert hall diffusion design? How useful is this approach?

To address these questions, a concert hall with a large upper volume (i.e., one in which the effects of reflectors were sizeable) was designed, and the internal sound field variations according to the reflectors and diffusers were examined using a GA simulation and 1:25 scale model. Furthermore, an optimal diffusion design was developed after comparing the diffusion effects according to various quantitative diffusion evaluation indices and acoustic parameters.

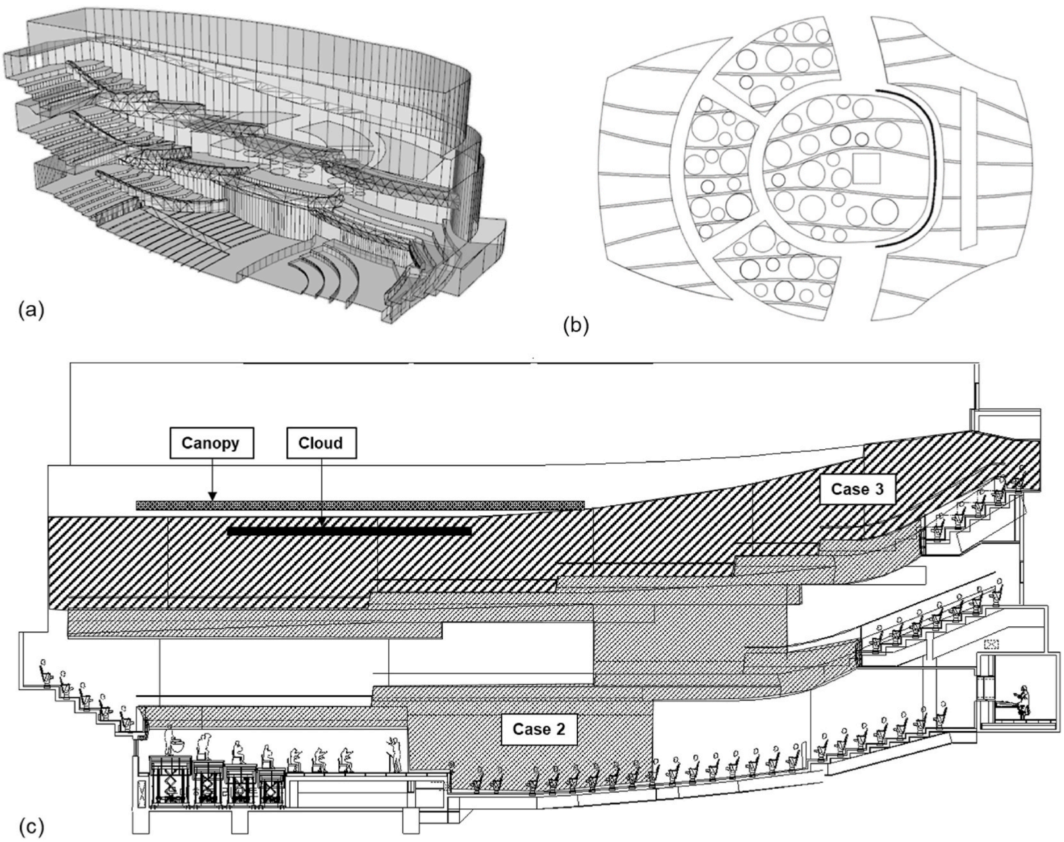

Fig. 1. Concert hall views: (a) perspective view, (b) plan of reflectors, and (c) section plan.

2. Methods

2.1. Concert hall acoustic design

2.1.1. Hall description

To propose an optimal design strategy for sound diffusion, a concert hall was designed in this study (Fig. 1). It was a symphony hall with an auditorium of 1450 seats and a total volume of 19 665 m^3 (V/N = 13.6 m^3). The height-to-width (H/W) and length-to-width ratios of the concert hall were 0.75 and 1.72, respectively. The hall was designed with countering walls featuring narrow widths, to ensure strong reflections from the side walls. In particular, a protruding structure for the side balconies on the second and third floors and a convex cross-sectional shape were adopted to produce these strong side reflections. To ensure rich reverberations and a sense of space inside the auditorium, abundant late reverberations resulting from the large volume around the upper part of the auditorium were provided (Fig. 1a). Six canopies were configured at the top of the stage to reinforce the sound pressure, and clouds of various sizes were added to induce appropriate reflections inside the space (Fig. 1b). In the canopy height range of 7–13 m, to ensure the proper ST range of −12 to −15 dB (considering the ensemble of performers on the stage), the canopy was fixed at a height of 11.5 m, considering the reflection paths to the auditorium [2]. The clouds were designed with diameters of 0.8, 1.2, and 1.6 m so that middle and high frequencies would be reflected appropriately inside the space.

2.1.2. Diffusion design

This paper proposes diffusion designs for five cases, depending on whether or not reflectors and diffusers were present. Table 1 outlines the characteristics of each case. The ratios of the diffuser area to the surface area of the entire wall and balcony front were also calculated and are listed in the table. In Cases 1–3 (which are classified according to the present or absence of diffusers), both a canopy and a cloud were installed above the stage; however, the diffuser locations and occupied areas differed: In Case 1, all the walls and balcony fronts were flat reflective surfaces, and a diffusion effect from the internal shape of the hall was observed; in Case 2, diffusers were attached to the side walls and balcony fronts on the first and second floors close to the sound source (excluding the stage), and the diffusion design was applied to 23% of the area of the entire wall and balcony front; in Case 3, diffusers were attached to the rear wall on the second floor and all walls on the third floor (which are far away from the sound source), and the diffusion design was applied to 34% of the area of the entire wall and balcony front. The optimal diffusion coverage density for 3D diffusers with a sectional height of 250 mm is 43% [42]. In this study, a coverage density of ~35% was selected, close to the optimal value required to achieve a uniform arrangement and facile diffuser installation. The canopies proposed in this study were designed to reinforce the sound pressure of the auditorium whilst considering the reflections of low-frequency sounds; meanwhile, the clouds were designed to ensure diffusion of the middle-to high-frequency bands. Hence, the canopies and clouds have different targets. Therefore, to investigate the additional diffusion effect under the variation in the concert hall acoustics (depending on the presence or absence of reflectors), we additionally selected Case 1B, wherein only canopies were installed (i.e., without clouds) and all the walls had flat reflective surfaces, and Case 1A, where both the canopies and clouds were removed.

2.2. Scale-model measurement

2.2.1. Diffuser profile

In the study, omnidirectional 3D diffusers with hemisphere-shaped profiles were used. The diffusers were composed of hard wooden materials with radii of 10 mm (1:25 scale factor and 250 mm structural height at full scale). Cypress wood was used as the diffuser material; it was sanded and finished with varnish four times for smooth surface treatment. To determine the physical acoustic properties of the diffusers, the absorption coefficient (Ac; ISO 354 [43]) and Sc (ISO 17497–1) were measured, as depicted in Fig. 2. All measurements were performed in a 1:10 reverberation chamber. Firstly, the Ac was measured using a spark source as the sound source, by employing four pin microphones (AKG C418) fixed at different heights. Measurements were performed twice with different sound source locations. The specimen area was set to 300 mm × 400 mm according to the ISO standard, diffusers were placed with the same coverage density as in the diffusion design, and a border was installed around the perimeter. Diffusers were attached to a wooden plate consisting of the same material as the wall in the real scale model. Separately, the sound absorption rate of the wooden wall material was measured. Next, the Sc was measured using a three-joint tweeter (AMT Mini8, Dayton Audio). To implement the diffusion sound field, the tweeter diaphragms were installed facing outwards at different corners inside the reverberation chamber. For the sound source, maximum length sequence signals were used. The same pin microphones (C418, AKG Acoustics) were used for the Ac measurements. Diffusers were placed at a coverage density of 35% in the diffusion design on a 420-mm-diameter acrylic disk located on a

Table 1
Diffusion design characteristics.

Cases	Reflectors		Diffuser location on wall						Surface area of wall and balcony		
	Canopy	Cloud	Balcony	1F	2F			3F	Diffusive [m^2]	Total [m^2]	Ratio [%]
				All	Front	Side	Rear	All			
1A									–	2251.7	–
1B	✓								–		–
1	✓	✓							–		–
2	✓	✓	✓	✓	✓	✓			588.8		23
3	✓	✓					✓	✓	902.3		34

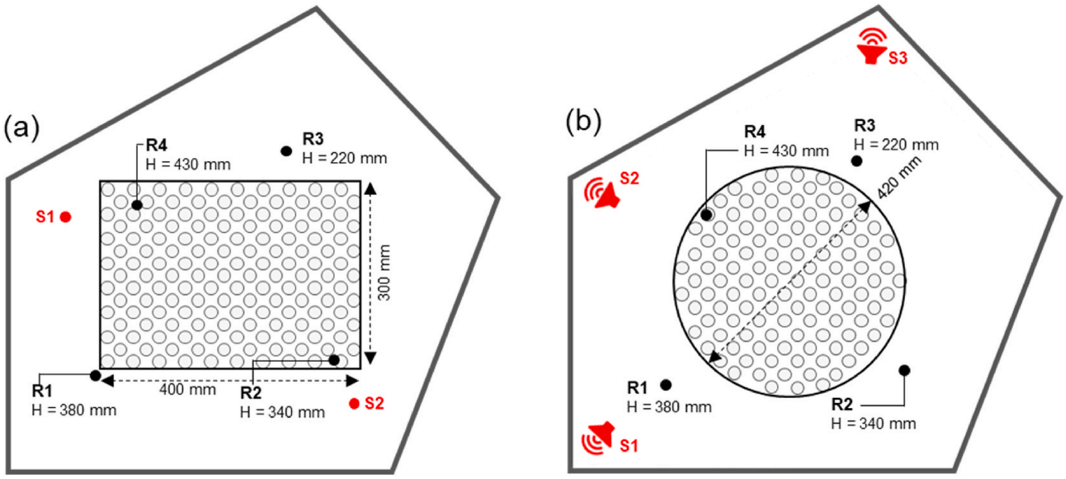

Fig. 2. Measurement setup in 1:10 reverberation chamber: (a) Ac and (b) Sc.

turntable, and a border was installed around the perimeter. Measurements were performed with the turntable in the fixed and rotating modes, with and without specimens. To apply the law of similarity, only the frequency band of 125 Hz–1 kHz was examined in this study at the real scale, considering the useable frequency ranges of the spark source and tweeter. The Ac was 0.07 and 0.012 on average for the wooden wall and diffuser, respectively (35% coverage density). Furthermore, the average Sc of the 3D diffuser was 0.25 in the 125 Hz–1 kHz band. This performance is similar to that achieved (with a coverage density of 43% and an identical diffuser) in a previous study [42], where the average Sc was 0.23 in the 125 Hz–1 kHz band.

2.2.2. Measurement conditions

The scale model was fabricated via 3D printing at a 1:25 scale, by laminating solid wood and then cutting the internal shape to maximally seal the interior from the outside. In a previous study [44], it was concluded that the 1:25 scale is appropriate when selecting the size and attachment locations of diffusers, and such scale models can incorporate both auditorium and stage acoustics for diffusion design. Therefore, a 1:25 scale model was also used in this study. The interior and reflector surfaces were varnished to form smooth reflective surfaces. Diffusers were attached to different inner surfaces according to the five diffusion designs and were uniformly arranged to produce a coverage density of 35% for the corresponding areas. Fig. 3a–c presents views of the stage and auditorium for each diffusion design case. The canopies were fabricated using the same wood material as that employed for the interior wall, and the clouds were separately produced using a 3D printer (Fig. 3d). All reflector surfaces were varnished in the same way. The auditorium seats and human (audience) models were fabricated similarly with an error of 0.05 or less compared to the sound absorption rate of a previous study [45]. For the auditorium seats, the sound absorption rate of light harvesting was used as a reference. A spark source and four 1/8 inch microphones with nose cones (GRAS 26 AC) were used as the source and receiver, respectively (Fig. 3e). The spark source matched that utilized to measure Ac. It exhibited omnidirectional and flat frequency characteristics in the 12.5–25 kHz band (500 Hz–1 kHz on the real scale). The sound source was placed at different positions according to the objective (i.e., to measure the stage and auditorium acoustics). First, when the stage acoustics were measured, the sound sources were positioned at four different points on the stage (S1–S4) at a height of 1.5 m, and the receivers were positioned at 1 m away in four directions at a height of 1 m. When measuring the auditorium acoustics, the sound source was fixed in the solo position at a height of 1.5 m, and the receivers were positioned at 20 points in the auditorium (nine positions on 1F, six on 2F, and five on 3F) at a height of 1.2 m (Fig. 3f).

2.3. Computer simulation

For the computer simulation, the GA simulation software Odeon (version 15) was used. It employs a hybrid calculation method that combines the image source method and early scattering method (for early reflections) with the ray-tracing method (for late reflections) [46]. To implement the diffusion sound field, the Odeon software employs the vector-base scattering model, which is based on a linear interpolation for specular and diffuse reflections. The scattered vector used here is realistic because it follows the random direction generated by the Lambert distribution. Regarding early reflections, secondary sources generated on each surface (according to the transition order) were reflected along with the image source, and the sound power was determined by the image and secondary sources distributed according to the Sc. Regarding late reflections, small secondary sources similar to the surface sources were generated when the late rays were reflected by the surface, and the oblique Lambert directivity (considering the roughness and edge diffraction of the surface) was reflected. Various other simulation tools are available, including CATT-Acoustic and Raven. However, when the changes in the acoustic parameters (with respect to Sc) were examined using these simulation tools in a previous study [24], only the Odeon simulation was able to partially indicate a difference exceeding the JND for the EDT. Therefore, the Odeon simulation software was utilized in this study, because it was considered to be the best tool for evaluating the diffusion sound field. The 3D concert hall modeling was performed in AutoCAD 2016 using the same shape as the scale model. The same values measured in the scale reverberation chamber were applied for the Ac and Sc of the interior finishing materials. In the Odeon simulation, the transition order was

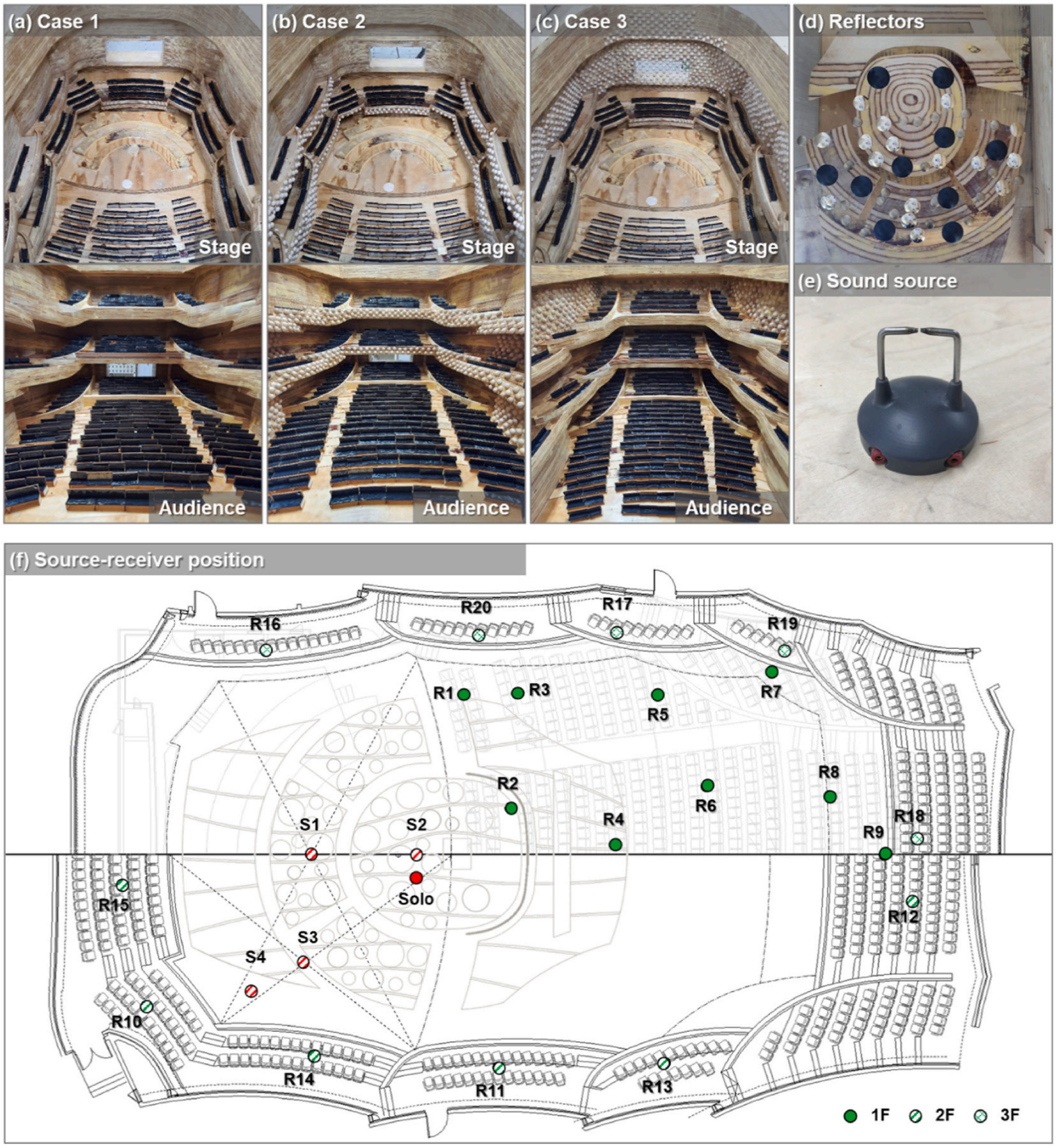

Fig. 3. 1:25 scale model measurement setups: (a) Case 1, (b) Case 2, (c) Case 3, (d) reflectors, (e) sound source (spark source), and (f) sound source and receiver positions.

set to seconds, and the impulse response length for calculating the indoor acoustic index was set to 5000 ms. In addition, 20 000 late rays were used; this is slightly higher than 13 448, the value recommended to facilitate mutual comparison with other diffusion designs. The locations of the sound sources and sound receiving points on the stage and in the auditorium were set to the same heights and positions as in the scale model. Regarding the sound source, a point sound source with omni-directional characteristics (similar to the spark source used in the scale model) was used, and the sound power of the entire frequency band was adjusted to 80 dB (the sound pressure level on the axis at 10 m was 49.0 dB) to ensure flat frequency characteristics in the 63 Hz–8k Hz band. Therefore, in this study, to thoroughly investigate the 500 Hz–1 kHz band, we treated the sound sources used in the scale model and computer simulation as having identical frequency characteristics. To verify the simulation, its results were compared with the measurements from the 1:25 scale model for Case 1A, which was taken as a reference. The simulation was found to agree well with the scale model, because the error was below 5% for the RT, EDT, and C_{80}.

2.4. Acoustic parameters

In the scale model, the changes in the concert hall acoustics with respect to the sound diffusion design were evaluated separately for the stage and auditorium. For the stage acoustics, the stage support parameters ST_{Early} and ST_{Late} were investigated. For the auditorium acoustics, the RT and EDT (indicating perceived reverberance) as well as C_{80} and the center time (T_S) (indicating the perceived clarity of sound) were evaluated. The RT was measured in an occupied auditorium, and the EDT, C_{80}, T_S, and ST were measured in an unoccupied one. Following ISO 3382–1 [32], the auditorium acoustic parameters were averaged for 500–1000 Hz, and the representative values were examined. The average value in the 250 Hz–2 kHz band is recommend for measuring ST. However, as described above, the average for 500–1000 Hz, which is the median of the 250 Hz–2 kHz band, was used as the representative value, considering the stability of each frequency band of the spark source employed in this study. Furthermore, the ST was split into ST_{Early} (for evaluating the ensemble according to ISO 3382–1) and ST_{Late} (for evaluating the perceived reverberance). When analyzing each acoustic parameter, the values were corrected considering the temperature and humidity at the measurement time. In addition, for a quantitative comparison of the diffusion effects according to the diffusion design, the Np and RSD of each acoustic parameter were analyzed [18]. In the case of computer simulation, the same calculation method as that employed in the scale model was also applied. ST_{Early} and ST_{Late} were evaluated as stage acoustics indices.

3. Results

3.1. Auditorium acoustics

3.1.1. Geometric acoustic simulation

Because the main objective of this study was to examine the changes in the acoustic parameters with respect to the diffusion design, the acoustic parameter change (Δ) was defined according to the values for Case 1, as shown in Eqs. (1) and (2). Based on this definition, Fig. 4 presents the GA simulation results for the source-to-receiver distance with respect to the sound diffusion design. The total average value of each case and the JND ranges of the acoustic parameters are indicated (ISO 3382–1 [32]). The standard specifies the JND as 5% for RT and EDT, 1 dB for C_{80}, and 10 ms for T_S.

In Case 1, the RT increased with the upper volume, because the reflectors were absent. In particular, the RT change due to the cloud was small (~2%), though the RT increase due to canopy removal was ~5%, higher than the JND. The RTs of Cases 2 and 3 were 3% and 4% lower on average, respectively, compared to that in Case 1. The RT decrease rate in Case 3 (with the large diffuser area) was high. The standard deviations (SDs) in Cases 2 and 3 were 3% and 5%, respectively, indicating a reduction effect exceeding the JND in some auditoriums. This is believed to be caused by the increase in total sound absorption, resulting from the increase in the inner surface area, itself attributable to the attachment of diffusers. The EDT shows a similar trend as the change in RT. The change in the initial

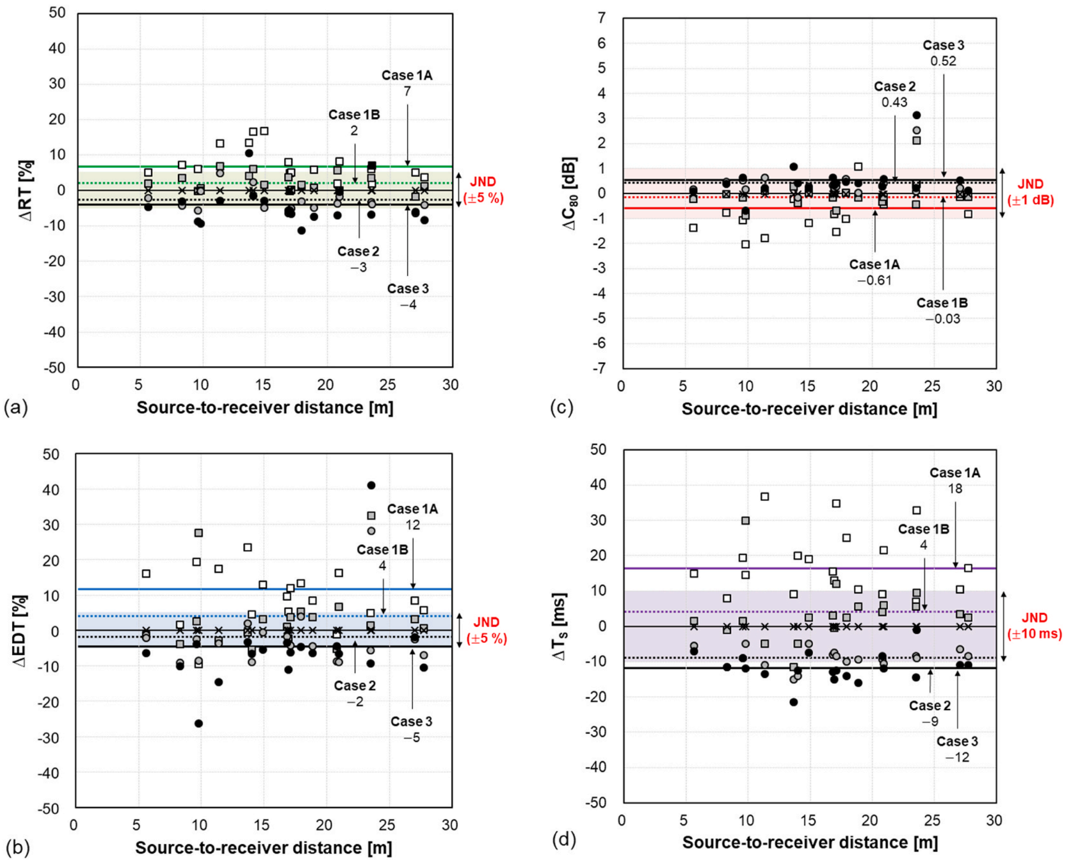

Fig. 4. Distribution of acoustic parameters in auditorium with respect to diffusion design in GA simulation: (a) RT, (b) EDT, (c) C_{80}, (d) T_S (□: Case 1A, ■: Case 1B, ×: Case 1, ●: Case 2, ●: Case 3).

decay energy attributable to the reflector was more conspicuous. The EDT change rates resulting from the removal of the cloud and canopy were 4% and 8%, respectively, showing large increases. Furthermore, considering that the SD for the EDT change with respect to the presence or absence of a cloud was ~10%, the cloud alone can produce differences that exceed the JND in certain auditoriums. In addition, the EDT was lower by 2% and 5% on average in Cases 2 and 3, respectively, compared to Case 1. In particular, Case 3 exhibits a decrease higher than the JND (5%). The cause of this difference is as follows. In Case 2, although sound absorption is achieved by the diffusers themselves, the diffusers attached to the walls around the stage and balcony front (i.e., those which are close to the sound source) increase the density of early reflections entering the auditorium. Hence, the energy reduction rate does not decrease significantly. However, in Case 3, diffusers are attached over a large area on the walls far from the sound source (including the walls on the third floor around the stage), which reduces both the early and late sound energies. Therefore, the influence upon the sound absorption of diffusers is dominant. C_{80} (the ratio between the early and late energies) shows an opposite trend to those of the RT and EDT, and the change is less than the JND (1 dB). In Cases 1A and 3, the SDs of C_{80} in the auditorium were 0.79 and 0.72 dB, respectively, suggesting that certain seats experience changes exceeding the JND. Therefore, installing a canopy plays a very significant role in strengthening the early sound pressure reinforcement, as partially confirmed by the RT and EDT. Meanwhile, according to the reduction trend of C_{80} in Case 3, the sound absorption effect of a diffuser attached far away appeared larger later, increasing the reduction rate of late energy compared to that of early energy. Next, the T_S (which indicates the perceived balance of reverberation and clarity) shows a trend similar to that of the reverberation parameters (RT and EDT). With the canopy and cloud absent, the T_S increased by up to 18 ms; then, when the diffuser was attached, it decreased by up to 12 ms. Furthermore, when only the impact of the cloud was considered, the average T_S increased by 4 ms when the cloud was removed. However, because the SD was 9 ms, differences exceeding the JND may arise in certain auditoriums. In other words, when the EDT is considered, the cloud significantly affects the early reverberations felt by the audience. Consequently, in the GA simulation, canopy removal significantly increases the RT, EDT, and T_S and decreases C_{80}, and cloud removal can increase the early reverberations. Furthermore, decreases in the EDT and T_S that exceeded the JND were more clearly observed in Case 3 (where diffusers were attached to a large area on the walls far from the sound source) than in Case 2 (where diffusers were attached to a small area on the walls and balcony front close to the sound source). The relevant expression are as follows:

$$\Delta RT[\%] \text{ and } \Delta EDT[\%] = \frac{Case\ X - Case\ 1}{Case\ 1}\ (X = 1A, 1B, 2, 3), \tag{1}$$

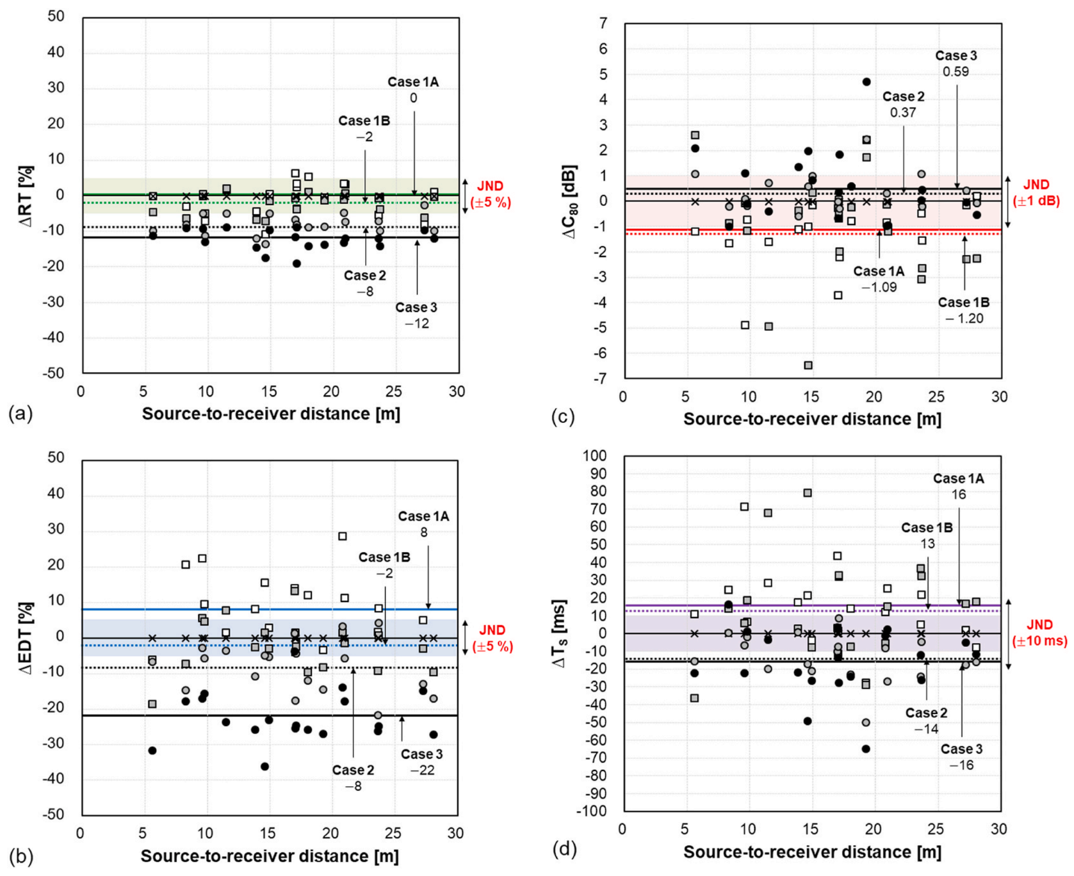

Fig. 5. Distribution of acoustic parameters in auditorium with respect to diffusion design in 1:25 scale model: (a) RT, (b) EDT, (c) C_{80}, (d) T_S (□: Case 1a, ■: Case 1b, × : Case 1, ●: Case 2, ●: Case 3).

$$\Delta C_{80}[dB] \text{ and } \Delta T_S[ms] = \text{ Case } X - \text{Case } 1\ (X = 1A, 1B, 2, 3). \tag{2}$$

3.1.2. Scale model

Fig. 5 presents the acoustic parameter results as functions of the source-to-receiver distance and with respect to the diffusion design in a 1:25 scale model. The change (Δ) of each acoustic parameter was calculated in the same way as in the GA simulation, and the average change and JND range are shown together. In the scale model, the change trend of each auditorium acoustic parameter resembled that in the GA simulation; however, the extent of change with respect to the diffusion design was greater. First, the RT did not exhibit a significant difference with respect to the presence or absence of reflectors. However, in Case 1A, the SD was 5%, confirming that the RT decreased significantly in certain parts of the auditorium, including the front rows, balcony seats around the stage, and choir seats, owing to the installation of canopies. The RT reduction rate due to diffuser attachment showed a similar trend to that in the GA simulation and was very large. In Cases 2 and 3, the average RT decreased by 8% and 12% on average, respectively, producing changes higher than the JND (5%). The EDT increase rate caused by the absence of a canopy was large (10%), as in the GA simulation. The difference produced by the absence of a cloud was 2%; however, considering that the SD was 7%, its effect on the EDT was significant compared with the JND. However, unlike the decrease of the EDT in certain auditoriums when a cloud was installed in the GA simulation, the EDT of the scale model increased in most auditoriums. Thus, the initial sound pressure reinforcement effect exceeded the effect produced by the cloud's reduction of the upper volume of the stage. This is believed to be caused by differences in shape. A flat cloud shape was used in the GA simulation because of the limitations of the modeling technology; meanwhile, in the scale model, the cloud was fabricated as a convex shape. Furthermore, the SD of the EDT was 10% when only a canopy was installed, though it decreased to 7% when a cloud was installed as well, suggesting that clouds can have a positive diffusion effect, such as producing a uniform EDT distribution in the auditorium. In Cases 2 and 3, where diffusers were attached inside the concert hall, the EDT decreased by 8% and 22% compared to Case 1, respectively, exhibiting changes much larger than the JND. The change in C_{80} due to canopy removal was small (at 0.11 dB), unlike in the GA simulation, and the decrease due to cloud removal was significant (at 1.20 dB), which was larger than the JND (1 dB). Interestingly, installing a cloud not only increased the EDT in the auditorium but also positively contributed to the increase in C_{80}. The average differences in C_{80} due to the installation of diffusers in Cases 2 and 3 were not large (at 0.37 and 0.59 dB, respectively); however, their SDs were high (at 0.74 and 1.42 dB, respectively). In particular, in Case 3, C_{80} increased close to the sound source, which seems to be an effect of sound absorption by the diffusers on the late reflections. Next, the T_S increased by 3 and 13 ms when the canopy and cloud were removed, respectively. Thus, the reverberations in the auditorium can be increased by a cloud. Therefore, as can be seen partially in the changes of EDT and C_{80} described above, a cloud can ensure clarity whilst increasing the initial perceived reverberance. In addition, T_S changed by −14 and −18 ms in Cases 2 and 3, respectively, because of the diffusers; these decreases exceed the JND (10 ms). To summarize, the main effect of installing a canopy is to reduce the concert hall upper volume, which reduces the RT in the auditorium near the stage and reduces the EDT and T_S in the auditorium overall. The initial sound pressure reinforcement effect of a cloud is more significant than the effect of the corresponding reduction of the upper volume, and C_{80} (as well as the EDT and T_S) increases. Essentially, diffusers decrease the RT, EDT, and T_S and increase C_{80} through the effect of sound absorption, regardless of attachment position and area.

3.2. Stage acoustics

The changes in stage acoustics produced by the reflectors and diffusers were classified into ST_{Early} and ST_{Late}. The average values at four points on the stage were calculated and are listed in Table 2. Here, ST_{Early} and ST_{Late} are the ratio of the energy at 100–1000 ms to the direct sound energies at 20–100 ms and 0–10 ms, respectively. In the GA simulation results, the changes in ST produced by the reflectors and diffusers were less than 1 dB. The ST does not have a proven JND, though Gade has estimated it to be ~2 dB [47]. Thus, the GA simulation did not yield any significant changes. In the results for the scale model, when a canopy was installed, strong early reflections were formed because the path of the primary reflection sound returning after hitting the ceiling became shorter. Consequently, strong early reflections were formed and ST_{Early} increased by 1.8 dB on average because of the sound energy reinforcement. In particular, at the director position (S2) (which extended forward from the center of the stage), the increase in ST_{Early} attributable to the installation of the canopy was very large (at 3.3 dB). It also increased by 1.5 dB and 2.0 dB at S1 and S3, respectively. The increase in ST_{Early} attributable to the cloud was insignificant, at 0.2 dB on average; however, some points (e.g., S1) showed increases of 1.4 dB. In contrast, ST_{Late} decreased by 0.4 dB because of the cloud and decreased by up to 1.9 dB at the director position (S2). Thus, the cloud not only reinforced the initial sound pressure but also reduced the reflections that returned to the stage after hitting the canopy, because it consisted of multiple convex sub-reflectors. The diffusers reduced both ST_{Early} and ST_{Late}. In Case 2, ST_{Early} and ST_{Late} decreased by 0.9 and 1.2 dB on average, respectively. ST_{Early} decreased the most (by 1.9 dB) at the stage center (S1), and ST_{Late} decreased by 1.6 dB at

Table 2
ST with respect to diffusion design.

Cases	Computer simulation		Scale model	
	ST_{Early} [dB]	ST_{Late} [dB]	ST_{Early} [dB]	ST_{Late} [dB]
1A	− 8.3	− 9.3	− 15.3	− 14.1
1B	− 9.1	− 9.0	− 13.5	− 13.4
1	− 8.4	− 9.2	− 13.3	− 13.8
2	− 8.5	− 9.7	− 14.2	− 15.0
3	− 8.3	− 9.8	− 14.6	− 15.5

S4, which is the closest point to the diffusers. In contrast, in Case 3, ST_{Early} and ST_{Late} decreased by 1.3 and 1.7 dB on average, respectively. At S1, ST_{Early} and ST_{Late} decreased the most, by 2.9 and 3.2 dB, respectively. Thus, in both Cases 2 and 3, the sound absorption effect was dominant. In Case 2, the ST was in the proper range of −12 to −15 dB and changed less than the JND (2 dB). Therefore, Case 2 was more appropriate than Case 3 in terms of the ST.

4. Discussion

4.1. Sound field distribution

According to the above descriptions, the diffusers exhibited significant sound absorption capacities, regardless of the area they occupied or their attachment positions. However, diffusers not only help solve acoustic defects such as tone coloration and false localization: they also even out the directional distribution of the reflections. Therefore, it is necessary to examine the distribution of acoustic parameters in the auditorium. To check the seat-to-seat deviations in the auditorium with respect to the diffusion design, the RSDs were calculated and are listed in Table 3. Because C_{80} includes negative values, it was excluded from the analysis. In addition, the results discussed above were summarized in Table 4, to examine the relationships between the RSD and other acoustic parameters. First, the RSDs of the RT and EDT among the seats decreased in both the GA simulation and scale model when diffuser attachments were installed. This trend was consistent with the decrease in the RSD of the auditorium EDT when diffusers were attached around the stage and side balcony soffits in opera houses of shoebox + horseshoe shapes in previous studies [18,26]. Furthermore, the RT and T_S showed different trends in the GA simulation and scale model. In the scale model, the seat-to-seat deviation in Cases 2 and 3 was less than that in Case 1, and the RT and T_S showed more uniform distributions than in Case 1. In contrast, the GA simulation showed lower uniformity in the RT and T_S in the auditorium. This difference also related to the fact that the GA simulation and scale model exhibited opposite results for the EDT change rate, as examined above. However, it is clear that the diffusers decreased the RSD of the EDT. However, there is a limitation in selecting the diffuser attachment position using only the RSD. Because the diffusers themselves have a very large sound absorption effect, the early and late energies can be excessively lost if the diffusers occupy a large area far from the stage, as in Case 3. Therefore, the study results suggest that attaching diffusers in a small area near the stage can reduce the EDT deviation among the seats without significantly reducing the RT and EDT, whilst also maintaining clarity. Next, to examine the effects of the reflector, the RSDs of the EDT and T_S in Case 1 (where both a canopy and cloud were installed) exceeded those of Case 1A, where neither a canopy nor a cloud were installed. In particular, considering both the GA simulation and scale model results, the uniformity of the EDT and T_S in both cases decreased significantly owing to the canopy. Because the canopy reduced the late reverberations and increased the early reflections entering the auditorium, the uniformity among seats of the long late reverberations could have decreased somewhat. Meanwhile, the positive diffusion effects of the clouds decreased the seat-to-seat deviations of the T_S and EDT. Therefore, installing a cloud evenly distributes the reflections entering the auditorium, and it improves the excessive reduction of EDT brought about by the canopy. When the stage support examined above was considered as well, the cloud had a positive effect on the stage acoustics because it slightly reinforced ST_{Early}. This result was partially consistent with the finding of Rindel [36], who reported that composing the reflectors as several pieces rather than as one large plane is advantageous for ensembles of musicians.

4.2. Effect of sound diffusion on number of reflections

To compare the diffusion effects quantitatively with respect to the diffusion design, Fig. 6 shows the Np for the impulse response of the auditorium in each case, as well as the averages. The Np was calculated as the sum of the number of all peaks within the range of −20 dB (based on the direct sound) and was divided into Np_{Early} and Np_{Late} at 80 ms on the time axis. First, the canopy increased both the early and late Np slightly, because the early reflections entering the auditorium after hitting the reflector increased. The diffusion effect produced by cloud installation increased Np_{Early}; however, Np_{Late} decreased by 21. Furthermore, after installing the canopy, the RSD of Np_{Early} remained the same (at 0.21), though after installing the cloud, it decreased to 0.19, thus reducing the seat-to-seat deviation. This result quantitatively verifies the initial diffusion effect of the cloud (as examined via the RSD_{EDT}) once again. The cloud increases C_{80} and decreases the T_S; this, combined with the Np_{Late} results, indicates that cloud installation increases the density of early reflections. Cases 2 and 3 show opposite results regarding the effects of diffuser installation with respect to the diffuser attachment position. In Case 2, where the diffusers were attached to the walls and balcony front close to the sound source, Np_{Early} increased significantly (by 19) owing to the increase in the density of early reflections by diffusion. Furthermore, Np_{Late} increased somewhat despite the sound absorption effect of the diffusers themselves upon late reverberations. Moreover, compared with Case 1, the RSDs for Np_{Early} and Np_{Late} were reduced by 0.01 and 0.03, respectively, in Case 2. In contrast, in Case 3 (where the diffusers were installed in a wide distribution on the wall far from the sound source), the sound absorption effect of the diffusers was dominant, as

Table 3
RSDs of acoustic parameters.

Cases	Computer simulation				Scale model			
	RT	EDT	C_{80}	T_S	RT	EDT	C_{80}	T_S
1A	0.04	0.07	–	0.19	0.05	0.11	–	0.17
1B	0.03	0.15	–	0.22	0.04	0.11	–	0.26
1	0.03	0.13	–	0.22	0.04	0.12	–	0.23
2	0.06	0.10	–	0.23	0.04	0.11	–	0.21
3	0.04	0.10	–	0.23	0.03	0.09	–	0.21

Table 4
Summary of differences in acoustic parameter changes (Δ) with respect to diffusion design. Bold values indicates that the change (Δ) exceeds the JND.

Cases		Auditorium acoustics				Stage acoustics	
		Δ RT [%]	Δ EDT [%]	Δ C_{80} [dB]	Δ T_S [ms]	ST_{Early}	ST_{Late}
1A	Simulation	**0.07**	**0.12**	−0.61	**18**	0.1	−0.1
	Scale model	0.00	**0.08**	**−1.09**	**16**	**−2.0**	−0.3
1B	Simulation	0.02	0.04	−0.03	4	−0.7	0.2
	Scale model	−0.02	−0.02	**−1.20**	**13**	−0.2	0.4
1	Simulation	Ref	Ref	Ref	Ref	Ref	Ref
	Scale model	Ref	Ref	Ref	Ref	Ref	Ref
2	Simulation	−0.03	−0.02	0.43	−9	−0.1	−0.5
	Scale model	**−0.08**	**−0.08**	0.37	**−14**	−0.9	−1.2
3	Simulation	−0.04	**−0.05**	0.52	**−12**	0.1	−0.6
	Scale model	**−0.12**	**−0.22**	0.59	**−16**	−1.3	−1.7

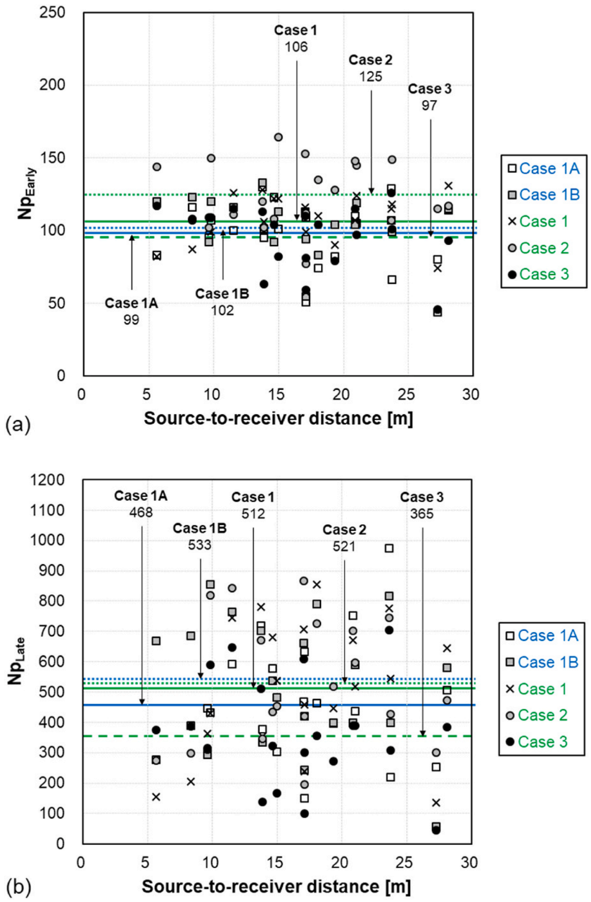

Fig. 6. Distribution of Np in auditorium with respect to diffusion design in 1:25 scale model: (a) Np_{Early} and (b) Np_{Late}.

shown in the RT and EDT. Both Np_{Early} and Np_{Late} exhibited their lowest values among all cases. In addition, compared with Case 1, the RSDs of Np_{Early} and Np_{Late} increased by 0.04 and 0.09, respectively, in Case 3, thereby increasing the seat-to-seat deviation. Consequently, Np demonstrates the positive diffusion effects of the canopy and cloud, and it confirms that the diffuser installation method in Case 2 is the most appropriate.

4.3. Influence of sound diffusion on energy decay curve

To examine the energy decay curve pattern with respect to the diffusion design, as well as to compare the decay rate variations over time, Fig. 7 shows the Schröder decay curve for each case. Fig. 7a presents the overall auditorium data and average data simultaneously, as well as the energy decay curves up to 2 s (based on the direct sound). Fig. 7b depicts the average energy decay curve up to

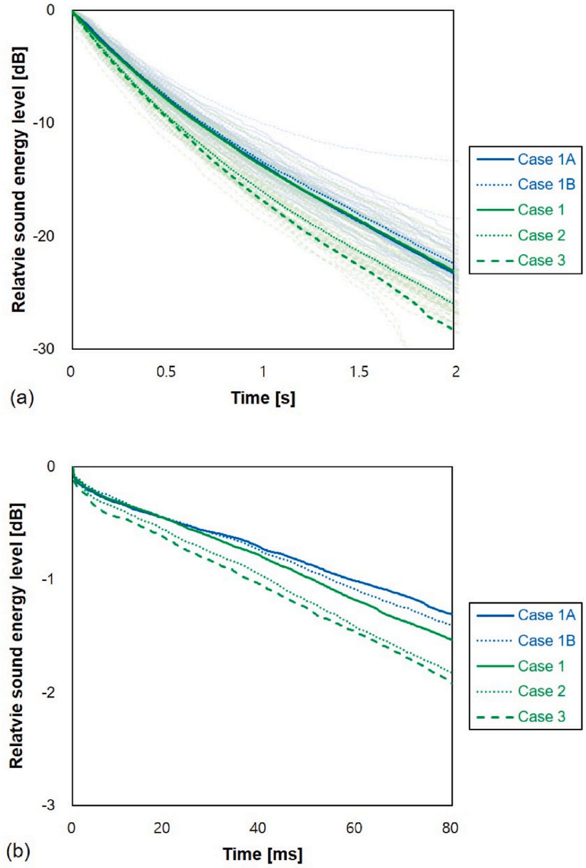

Fig. 7. Schröder decay curves with respect to diffusion design in 1:25 scale model: (a) for all receivers at 0–2 s and (b) mean values at 0–80 ms.

80 ms, to demonstrate the variations in early reflections. The general energy decay pattern exhibited a trend similar to that of the RT. The early section demonstrates that adding the canopy increases the energy until ~20 ms, with a gentle curve shape, though the slope of the decay curve increased after 20 ms. However, with a cloud installed as well, the slope was gentle for the first 20 ms but decreased toward the later stage, similar to Case 1A. This trend appears to be due to an increase in the EDT, as described above. Meanwhile, both Cases 2 and 3 were significantly affected by the diffuser sound absorption. In Case 3, where the diffusers were installed near the back rows (which are significantly affected by late reflections), the reduction rate of the decay curve was larger. However, Case 2 presents a gentler curve than Case 1 and an attenuating pattern toward the later stage. In contrast, Case 3 shows sharp energy attenuation from the early stage, and the energy reduction rate increases toward the later stage. This tendency arises because the early energy ratio increases somewhat under the effect of the early reflections generated by the diffusers installed around the stage where the direct sound reached, as shown in the Np results. This effect can be confirmed by the reductions of the RT, EDT, and T_s in the auditorium, as described above. However, it must be considered that installing diffusers reduces the deviations of the parameters among the seats. Therefore, it is necessary to find an appropriate tradeoff relationship between the absorption and diffusion effects of diffusers. Using diffusers excessively can sharply reduce the reverberations experienced by the audience in the auditorium, by decreasing not only the reverberations (via sound absorption) but also Np. However, installing an appropriate level of diffusers can increase intimacy without significantly damaging the reverberations in the concert hall auditorium, by increasing Np within a range that does not significantly hamper the reverberations [11].

4.4. Usability of scale model for diffusion design

In general, various methods (from scale models to complex simulation models) are used to predict the acoustic properties of a space [48]. The scale model is known to provide sufficiently accurate results for predicting the sound of an actual space using small microphones and speakers, under the assumption of the law of similarity. Even today, scale models in various scales are used in high-cost concert hall projects. However, scale models involve high costs, a long production period, and considerable personnel. Therefore, computer simulation have been utilized to replace them. Theoretically, the finite element method or boundary element method, based on three-dimensional wave equations, are the most accurate methods for predicting the indoor sound field; however, they require the input of complex input conditions, such as boundary conditions. GA software, which models sound waves as rays, simplifies the physical behavioral properties of sound and is very efficient in explaining the geometrical structure of a space and the distribution of surface material properties [48]. GA software assumes a point sound source and calculates the energy that reaches the receiving point from the sound source (via direct transmission or reflections on internal surfaces) by capturing acoustic phenomena such as scattering and sound absorption. Various calculation methods [e.g., the image source method (ISM) and ray-tracing method (RTM)] have been applied to GA software, and hybrid-method-based tools that combine various methods (like the Odeon software used in this study) have been commercially applied. However, ISM ideally requires an infinite reflecting surface, and RTM has a low spatial and temporal resolution due to the analysis being conducted in the energy domain; hence, the GA software is still limited in terms of completely replacing the scale model.

In this study, the overall trends of the scale model and GA simulation results were similar, though the changes in the GA simulation results were smaller. Further, the EDT changes produced by clouds in the auditorium differed between the two sets of results. There are various reasons for this difference, as follows. First, the precision may differ because, unlike the scale model, the GA simulation simplifies the internal shape in the 3D modeling process [49,50]. Furthermore, owing to the nature of the geometrical acoustics software (which is based on Lambert scattering models), the edge diffraction of small facets may be neglected to some extent [51]. In addition, the geometrical characteristics of the actual diffuser are not reflected, because in the simulation, numerical corrections were performed only by applying Sc to the flat surface, rather than modeling a diffuser that corresponded exactly to the real shape [23]. Consequently, the changes in auditorium acoustics after applying the diffuser were not significant in the GA simulation results. In a previous study [11], a diffusion effect was, by performing measurements using a 1:10 scale model, observed in one case that was not found in GA simulations. However, the overall trends of the acoustic changes in the GA simulation and scale model were in agreement. Therefore, GA simulation is useful for predicting the overall trends, though it is more appropriate to use a scale model to examine the detailed and complicated diffusion effects of reflectors and diffusers when optimizing concert hall designs.

4.5. Comparison with previous studies

In this study, it was found that placing diffusers close to the sound source makes it possible to provide uniform sound to the auditorium, by reducing the RSD whilst also minimizing the RT, EDT, and T_S reduction; it also provides rich sound by increasing Np. It was also shown that the ST can be suitably distributed in terms of stage acoustics when diffusers are applied appropriately. In previous studies [25–28], the RT was found to decrease when diffusers were installed in a concert hall, and similar results were observed in this study. However, the effect of C_{80} was found to differ according to the auditorium location. Ryu & Jeon [26] and Jeon et al. [11] presented somewhat opposite results, in that the C_{80} of the auditorium near the stage decreased when diffusers were installed. This appears to be due to the difference in the internal geometry and volume of the concert hall. While previous studies focused on analyzing the meaning of the acoustic parameter results by seat, to investigate the distribution between auditoriums, this study was significant in that the distribution of acoustic parameters between auditoriums was quantified using the RSD index.

In terms of the effects of reflectors, it was found that canopies increase the ST but decrease the RT or EDT, and that clouds have a complementary effect to canopies because they increase the EDT and C_{80}. Previous studies [33,34], and [36–38] examined the effects of stage design factors on stage acoustics, and confirmed the ST increase effect via stage diffusion design factors, as performed in this study. However, few studies have been conducted regarding the influence of such stage design factors on the auditorium. Several studies [35] considered the loudness of the auditorium as well, though almost no studies have evaluated the effects of stage design factors on the auditorium sound field using various acoustic parameters, as we performed here. Therefore, we believe that this study proposes a new approach, by investigating the internal sound field changes attributable to concert hall diffusion design considering both auditorium and stage acoustics. Moreover, whilst most of the previous studies selected either a scale model or a computer simulation model, both methodologies were applied and examined in this study. Therefore, the results of this study are expected to help architects and sound consultants select appropriate methodologies for concert hall diffusion designs in the future.

4.6. Limitation and further research

This study has the following limitations. First, it proposed a diffusion design methodology for a rectangular hall, and hence further research should investigate its application to halls of various shapes in the future. In addition, the sound field change due to diffusion design was investigated for the 500–1000 Hz frequency band, though a frequency band of 1 kHz or higher can be applied as a target for diffusers, depending on the purpose. Therefore, further research is required on ranges besides the frequency band examined in this study. In addition, diffusion designs were here presented according to physical indices, though audibility evaluation must also be considered from the perspective of human hearing.

5. Conclusions

In this study, the variations of stage and auditorium acoustics with respect to the installation of reflectors (canopies and clouds) and diffusers in a concert hall were investigated, and guidelines for optimal diffusion design were presented. To this end, the differences in sound field variations with respect to the diffusion design were examined (based on the JND) using a computer simulation and 1:25 scale model. The simulation and scale model produced similar overall trends, though the scale model showed clearer differences in the effects of the diffusion design. The main findings of this study were as follows:

- The canopies reduced the RT of the auditorium near the stage and reduced the EDT and T_S in most of the auditorium, by reducing the upper volume. Furthermore, they increased the ST_{Early} and ST_{Late} by generating strong early reflections. The clouds increased the EDT and C_{80} in the auditorium because they were composed of multiple sub-reflectors, resulting in a larger capacity to reinforce the early sound pressure than that achievable by reducing the upper volume. Furthermore, the clouds reduced the deviations of the

EDT and T_S produced by canopy installation in the auditorium. Moreover, the clouds partially reduced the ST_{Late} via the diffusion effect, though they partially increased the ST_{Early}. Therefore, the effects of clouds complement those of canopies.
- The diffusers decreased the RT, EDT, and T_S more than the JND when they were attached to a small area on the walls and balcony front near the stage (Case 2) and when they were attached to a large area on the rear walls of the auditorium and the entire wall on the third floor (Case 3). Furthermore, C_{80} increased in some of the auditorium. However, the ST decreased more than the JND in Case 3 only. The diffusers decreased the seat-to-seat deviations of the acoustic parameters, regardless of their attachment positions. However, when Np and the energy decay curve were considered comprehensively, attaching diffusers to the protruded surfaces near the sound source (as in Case 2) can be considered the optimal design for minimizing reverberation reduction in the auditorium and reducing seat-to-seat deviations whilst increasing the density of early reflections. Case 3 also reduced the seat-to-seat deviations to some extent, though it was somewhat inappropriate in the concert hall investigated in this study because it significantly reduced the reverberations in the auditorium. Therefore, diffusers should be designed to maximize diffusion whilst minimizing sound absorption, by considering the tradeoff between the sound absorption and diffusion.
- The GA software includes aspects that can be improved, because when the computer simulation results were compared with those of the 1:25 scale model, only the latter showed a positive diffusion effect from the cloud, clear ST change pattern, or clear quantitative differences in the auditorium acoustic parameters. Thus, it is desirable to use the scale model alongside a computer simulation in the early stages of diffusion design.

The generalizability of the results of this study may be limited because only one concert hall was considered and only the changes in the mid-frequency band were examined. Furthermore, owing to the functional limitations of the spark source used in this study, the effect of the diffusion design was examined only for the 500–1000 Hz band. Nevertheless, this study is significant in that diffusion design factors (e.g., reflectors and diffusers) for a concert hall were (for the first time) examined by comprehensively considering the auditorium and stage acoustics. The results of this study could be used to inform the diffusion designs of future concert halls.

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Author statement

We confirm that there are no known conflicts of interest associated with this publication and that this work's financial support did not influence its outcome. We also confirm that the manuscript has been read and approved by all named authors, and that there are no unlisted persons who may have also satisfied the criteria for authorship. We have approved the order of authors listed in the manuscript.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. As such, we affirm that we have followed the regulations of our institutions concerning intellectual property.

We further confirm that any aspects of the study that involve humans have received ethical approval by all relevant bodies and that such approvals are acknowledged within the manuscript.

We understand that the corresponding author is the sole point of contact for the editorial process (including correspondence with the editorial manager and direct communications with the office). Jin Yong Jeon is responsible for communicating with the other author regarding the publication progress, revision submissions, and final approvals of proofs. We confirm that we have provided a current and correct email address that is accessible to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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